

Multiple Choice

1. **Answer:** C

Options: A) True, True; B) False, False; C) True, False; D) False, True

Note: The correct answer is true for Statement 1 because every integral domain allows the construction of a field of quotients that includes all fractions formed from its elements, while Statement 2 is false because a polynomial of degree n can have fewer than n zeros or infinitely many zeros if it is not a polynomial over a field (for instance, in the ring of integers).

2. **Answer:** A

Options: A) True, True; B) False, False; C) True, False; D) False, True

Note: Both statements are correct because a homomorphism preserves the group structure, ensuring that the image of a cyclic group remains cyclic and the image of an Abelian group retains the commutative property.

3. **Answer:** A

Options: A) 4; B) 7; C) 8; D) 12

Note: The value of n is 4 because there are exactly two abelian groups of order 4, which are the cyclic group of order 4 ($\mathbb{Z}/4\mathbb{Z}$) and the direct product of two cyclic groups of order 2 ($\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$), according to the Fundamental Theorem of Finite Abelian Groups.

4. **Answer:** C

Options: A) local maximum at P and at Q; B) saddle point at P and at Q; C) local maximum at P and a saddle point at Q; D) local minimum at P and a saddle point at Q

Note: The function $f(x, y) = x^3 + y^3 + 3xy$ exhibits a local maximum at point P and a saddle point at point Q due to the critical points found through setting the gradient to zero, which reveals distinct characteristics of the Hessian matrix's eigenvalues at these points.

5. **Answer:** A

Options: A) 6; B) 12; C) 30; D) 105

Note: The maximum possible order of an element in the symmetric group S_n is equal to the least common multiple of the sizes of the cycles in its disjoint cycle decomposition, and for $n = 6$, the maximum occurs with a 6-cycle, which has an order of 6.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: The polynomial $4x - 2$ is irreducible over the field of rational numbers $\mathbb{Q}$ because it is a linear polynomial with no rational roots, making it unable to be factored further into polynomials of lower degree with rational coefficients.

7. **Answer:** False

Options: A) True; B) False

Note: The statement is false because the degree of the product of two polynomials $f(x)$ and $g(x)$ is equal to the sum of their degrees, not the maximum of their degrees, specifically when both polynomials are non-zero.

8. **Answer:** False

Options: A) True; B) False

Note: The area of an equilateral triangle can be calculated using the formula $A = r * s$, where r is the radius of the inscribed circle and s is the semi-perimeter, and for a triangle with an inscribed circle of radius 2, the area is actually 12, not 16.

9. **Answer:** False

Options: A) True; B) False

Note: There are only two homomorphisms from $\mathbb{Z}$ to $\mathbb{Z}_2$, corresponding to the images of 0 and 1 in $\mathbb{Z}_2$, since any homomorphism must map 1 in $\mathbb{Z}$ to an element in $\mathbb{Z}_2$, and the only elements in $\mathbb{Z}_2$ are 0 and 1, leading to a finite number of distinct homomorphisms.

10. **Answer:** True

Options: A) True; B) False

Note: The statement is true because the integers 3, 5, 7, and 11 are the only prime numbers, and for each prime number n , there exists exactly one group of order n , which is the cyclic group of order n , making them unique up to isomorphism.

Long Form Response

11. **Answer:** The statement "Every solution is a scalar multiple of $(-5, 1, 1, 0)$ " is true if the given system of linear equations has a solution space that is one-dimensional and spanned by the vector $(-5, 1, 1, 0)$. This occurs when the system is consistent and has infinitely many solutions, specifically if the equations are dependent. In such cases, all solutions can be expressed as scalar multiples of a single vector, indicating that the solution set forms a line through the origin in the direction of $(-5, 1, 1, 0)$. Thus, if the analysis of the system confirms that it is indeed structured in this way, the statement holds true.

Note: correct because if the system of linear equations is consistent and dependent, it results in a one-dimensional solution space spanned by the vector $(-5, 1, 1, 0)$, meaning all solutions can be expressed as scalar multiples of that vector.

12. **Answer:** When two $n \times n$ invertible matrices A and B are similar, it implies that there exists an invertible matrix P such that $B = P^{-1}AP$. This similarity has several implications: 1. The matrix $A - 2I$ is similar to $B - 2I$, which means they share the same eigenvalues, and thus their spectral properties are preserved. 2. The trace of A , defined as the sum of its eigenvalues, is equal to the trace of B , reflecting that similar matrices have the same trace regardless of their specific entries. 3. The inverses A^{-1} and B^{-1} are also similar, as $B^{-1} = P^{-1}A^{-1}P$, demonstrating that the relationship extends to their inverses as well. Thus, the similarity of A and B fundamentally influences not only their direct properties but also those of derived matrices like $A - 2I$, their traces, and their inverses.

Note: correct because it accurately describes how the similarity of two invertible matrices A and B ensures that their transformations, spectral properties, traces, and inverses are related in a consistent manner, demonstrating that similar matrices maintain key characteristics such as eigenvalues and trace.